

Experiment 'p_5_9_a_sphere_volume' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_5_9_a_sphere_volume

Natural language statement: Write a method that computes the volume of a sphere given its radius.

Method signature: p_5_9_a_sphere_volume(r: real) returns (volume: real)

Ensures

- `volume == (4.0 / 3.0) * 3.14159 * r * r * r`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that computes the volume of a sphere given its radius.

The signature should be:

`p_5_9_a_sphere_volume(r: real) returns (volume: real)`

The method should respect the following contract:

`ensures volume == (4.0 / 3.0) * 3.14159 * r * r * r`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmzHPVbYwRASvOFIovKzBnTjz1EZ6

Model: gpt-5.5-2026-04-23

Created at: 1780567675

Finish reason: stop

Usage: 296 tokens in, and 134 tokens out

```
{  
  volume := (4.0 / 3.0) * 3.14159 * r * r * r;  
}
```

Final Program

```
method p_5_9_a_sphere_volume(r: real) returns (volume: real)  
  ensures volume == (4.0 / 3.0) * 3.14159 * r * r * r  
{  
  volume := (4.0 / 3.0) * 3.14159 * r * r * r;  
}
```

Total Token Usage

Input tokens: 296

Output tokens: 134

Reasoning tokens: 87

Sum of ‘total tokens’: 430

Experiment Timings

Overall Experiment started at 1780567674436, ended at 1780567715461, lasting 41025ms (41.03 seconds)

Iteration #1 started at 1780567674437, ended at 1780567715461, lasting 41024ms (41.02 seconds)